LeapTo's Forecasting Model: Methods, Results, and Validation

Oct 24, 2026

Summary

LeapTo warns city health offices when leptospirosis risk is rising after heavy rain. The data was trained from at least 6 years of data and in the nine years it never saw, it ranks a city's riskier weeks correctly about 77% of the time versus 67% for a seasonal calendar.

- **Data:** 8,034 leptospirosis cases in 12 Philippine cities, 2008–2022, matched week by week to NASA IMERG satellite rainfall.

- **Timing:** risk peaks **2 weeks after a heavy-rain week**, at 1.77× a typical week, and stays raised for about 5 weeks.

- **Size:** one heavy-rain week (163.8 mm, the wettest 5% of all city-weeks) adds about 2.5 typical weeks' worth of cases. Each city's own wettest 5% of weeks account for about 23% of all cases.

- **Forecasting:** the final model forecasts 1–2 weeks ahead. It beats the seasonal calendar, a "same as last week" rule, and our earlier model in a strict rolling test.

- **Case studies:** the app would have warned of the Navotas 2018 outbreak 2–3 weeks before case reports, but not of Typhoon Washi; for such floods, the app adds a separate Flood alert that the health office switches on.

- **Flooding:** cases cluster in flood-prone barangays, heavy rain is more dangerous where more people live in flood-hazard areas, and a rising river raises risk for about 8 weeks. Neither improves the actual forecasts, so they explain the risk rather than predict it. ***For the app implementation, we can list down the flood prone barangays per city so that LGU's can allocate more resources to them.***

- **Limitations:** weekly case *numbers* cannot be forecast precisely, because most city-weeks have 0–2 cases. The app therefore shows **risk levels**, not case counts.

Data

Everything is weekly, on the Department of Health's morbidity-week calendar, for 12 cities from 2008 to 2022 (9,360 city-weeks).

Data	Source	Detail
Leptospirosis cases	DOH PIDSR, via Project CCHAIN	Weekly counts per city, 2008–2022. Iloilo has the most (3,948); Mandaue the fewest (39) (Thinking Machines Data Science, 2024).
Rainfall (main)	NASA IMERG Final, 0.1° satellite grid	Daily, 2003–2022, averaged over each city's barangays weighted by population (Huffman et al., 2023a)
Rainfall (real-time)	NASA IMERG Late	Available within a day; used to test forecasting as the app would run it (Huffman et al., 2023b)
Rainfall (check)	CHIRPS, via Project CCHAIN	Used to compare sources; it underestimated extreme weeks such as Typhoon Washi (Funk et al., 2015)
Population	WorldPop, via Project CCHAIN	Barangay population per year; 2021–2022 extended from the trend
Flood hazard	UP NOAH maps, via Project CCHAIN	Share of each barangay's area in low, medium or high hazard for 5-, 25- and 100-year floods

Why IMERG over CHIRPS: in a model screening test, IMERG Final explained cases better than CHIRPS (AIC −141 vs −78 against no rain), and it captured Typhoon Washi's rainfall, which CHIRPS largely missed.

Data-quality rules, applied before modelling:

- Long runs of zero-case weeks that are unlikely by chance are treated as non-reporting and left out. The rule adapts to each city's usual case rate.
- City-years with unusually low reporting of other diseases too are left out (pre-COVID only).
- Weeks from 15 March 2020 are flagged as the COVID period, when reporting dropped.
- In total, 386 of 9,360 city-weeks (4%) are excluded from fitting.

The Model and its Settings

LeapTo uses one model family, a negative binomial **distributed lag non-linear model (DLNM)**, in two versions: one to *explain* how rain affects risk, one to *forecast* it (Gasparrini, 2011).

Why this model:

- **Negative binomial:** weekly counts are far more variable than a Poisson model allows (overdispersion confirmed in exploratory analysis).
- **Distributed lag:** rain does not cause cases the same week. It raises risk over the following weeks, and the model estimates that delay rather than assuming it.
- **Non-linear:** a little extra rain matters less than a flood-level week, so the rain effect is a curve, not a straight line.
- **One pooled model with city effects:** small cities borrow strength from large ones, while each city keeps its own level, season and dispersion.

Explanatory model (all years, used for the rain–risk findings):

$$\log \mathrm{E}[Y_{ct}] = \log\left(\mathrm{pop}_{ct} \cdot \tfrac{d_t}{7}\right) + \alpha_c + s_c(\mathrm{week}) + f_c(\mathrm{trend}) + \beta\,\mathrm{COVID}_t + \sum_{\ell=0}^{10} w(\mathrm{rain}_{c,t-\ell},\, \ell)$$

Forecasting model (final version, h = 1–4 weeks ahead):

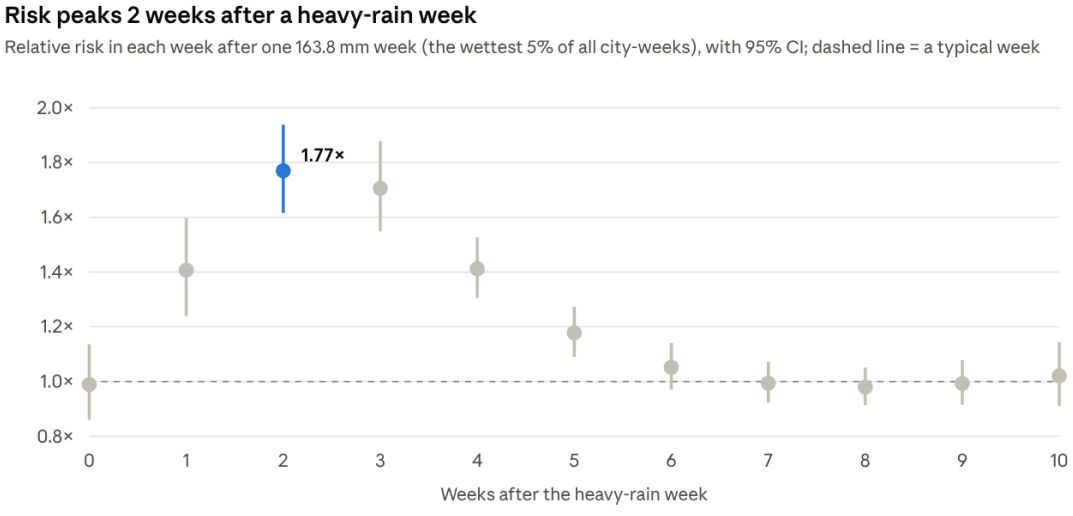
$$\log \mu_{c,t+h} = \mathrm{offset} + \alpha_c + s_c(\mathrm{week}) + \gamma_1 \log(1 + Y_{c,t-1}) + \gamma_2 \log(1 + \bar{Y}_{c,t-4:t-2}) + \delta \log(\mathrm{rate}_{c,\,52\ \mathrm{wk}}) + \beta\,\mathrm{COVID} + \sum_{\ell=h}^{10} w(\mathrm{rain}_{c,t+h-\ell},\, \ell)$$

Setting	Choice	Why
Outcome	Weekly cases, offset by population and week length	Rates comparable across cities; DOH week 1 and week 52 can be 4–12 days long
Rain measure	Weekly rainfall, IMERG (Final to explain, Late to forecast)	Best-performing source; Late is available in real time
Rain curve	Natural spline, knots at the 25th, 50th and 75th percentiles	Chosen by AIC among 9 options
Lags	0–10 weeks, natural spline with 3 knots on a log scale	10 weeks chosen by AIC; covers the incubation period plus reporting
Seasonality	3 sine–cosine pairs per city	Captures each city's rainy-season shape
Trend (explanatory)	Natural spline per city, 7 df	Long-term change in reporting and risk
Recent level (forecast)	Each city's case rate over the last 52 known weeks	Replaced the fitted trend: the biggest single forecasting improvement

Setting	Choice	Why
Case history (forecast)	Last known week, plus the mean of the 3 weeks before it	Outbreaks build over several weeks
Typhoon Washi (forecast)	Washi's 5 outbreak weeks in Cagayan de Oro set aside as training outcomes; they still count as recent case history	One event is 73% of the city's cases; otherwise the model expects a Washi-sized outbreak every December–January
Reporting delay	Cases assumed known 1 week late	Matches real PIDSR reporting
Dispersion	One per city; the 3 smallest cities share one	Stable estimates for cities with under 70 cases
Reference	A typical week (pooled median, 35.6 mm)	Risks read as "compared with a normal week"
Software	R: `glmmTMB`, `dlnm`	Standard tools for this kind of model (Brooks et al., 2017; Gasparrini, 2011; R Core Team, 2024).

Model Outcomes

Heavy rain raises leptospirosis risk with a clear delay: little the same week, a peak 2 weeks later, and a return to normal after about 5–6 weeks.



Explanatory model, IMERG Final, 12 cities, 2008–2022

The delay fits the disease: an incubation period of about 1–2 weeks after contact with contaminated floodwater, plus time to fall ill, seek care and be reported. The same 2–3 week peak appeared under all three rainfall sources.

Rain in one week	Weekly rainfall	Risk in the peak week	Extra cases over the next 10 weeks
Wettest 25% of all city-weeks	73.8 mm	1.14× (week 3)	0.6 typical weeks' worth
Wettest 10%	121.5 mm	1.44× (week 2)	1.5
Wettest 5% (heavy rain)	**163.8 mm**	**1.77× (week 2)**	**2.5 (95% CI 1.9–3.3)**
Wettest 1%	265.2 mm	2.97× (week 2)	5.7

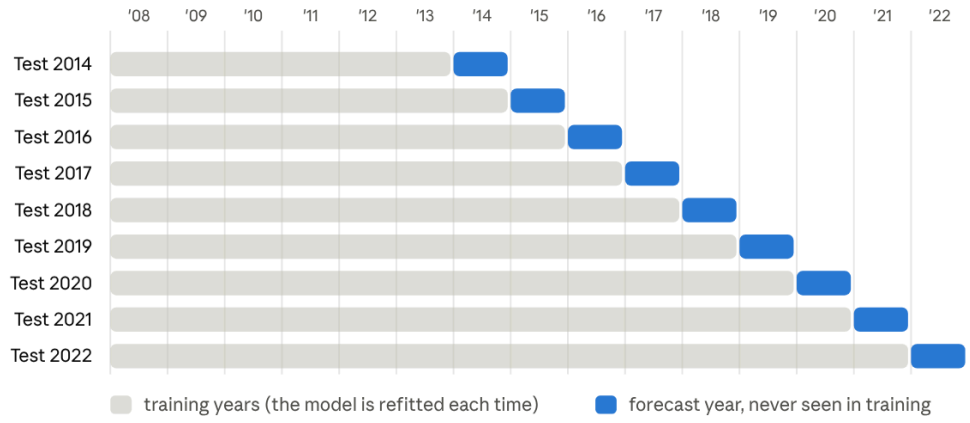
Cases attributable to rain: about **23%** of all cases (95% CI 21–26%) are attributable to heavy-rain weeks (each city's own wettest 5%), and about 47% to any rain above a typical week. The share is highest in Dagupan (56%), Muntinlupa (51%) and Navotas (48%), and lowest in Zamboanga (8%) and Davao (12%).

Epidemic thresholds: following the DOH rule, a week is an outbreak week when cases exceed the mean plus 2 standard deviations of the same week (± 2 weeks) in the previous 5 years. A version that leaves past outbreaks out of the baseline flagged 6.8% of city-weeks, against 3.6% for the standard rule. (National Epidemiology Center, 2014, p. 57). The standard rule stays inflated for years after a large outbreak: in Iloilo it flagged no weeks at all in 2013–2016. LeapTo uses the outbreak-excluded version.

Validation Settings

The forecasts were tested the way the app would run: each year was forecast using only the data that would have existed at the time, and scored against what actually happened.

Each year from 2014 to 2022 was forecast using only earlier years



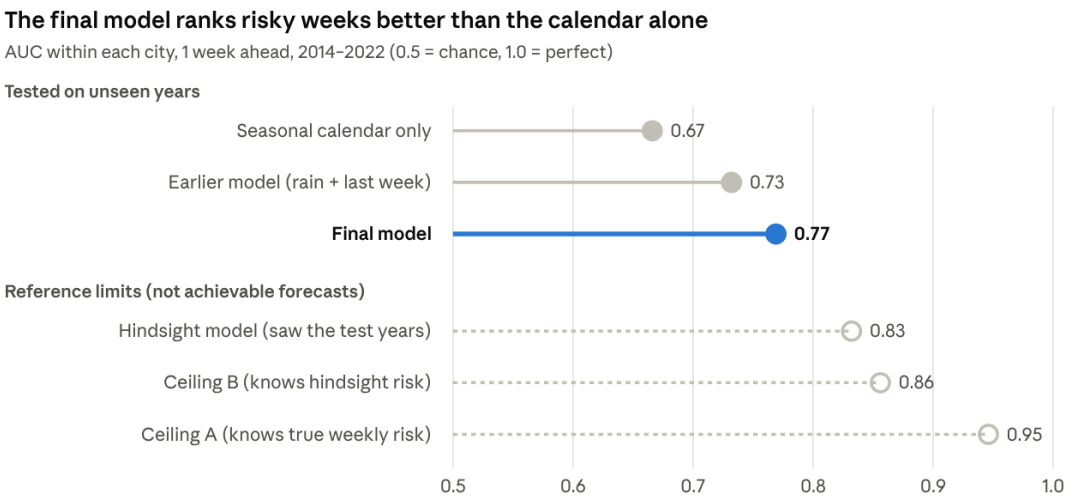
Rolling-origin validation · 9 test years · 12 cities

Setting	Choice
Design	Rolling origin: train on 2008 to year Y, forecast every week of Y+1, for 9 test years (2014–2022)
Horizons	1, 2, 3 and 4 weeks ahead, one model per horizon, using only rain that has already fallen
Reporting delay	Cases assumed known 1 week late, as in real PIDSR reporting
Rainfall	IMERG Late for training and forecasting (what the app can get in real time). Checked against Final → Final and Final → Late
Comparison models	Seasonal calendar only ("July is usually bad here"); persistence ("same as the last known week"); the earlier Stage 5 model
Outbreak definition	A week above the DOH epidemic threshold (5-year, outbreak-excluded) (National Epidemiology Center, 2014).
Scores	**Log score** (the main criterion: how much probability the forecast gave to what happened), **AUC within each city** (how well it ranks risky weeks), **precision and sensitivity** of alerts at fixed cut-offs, false alarms per city per year, and % error of weekly and yearly case totals
Uncertainty	Differences between models tested by a block bootstrap over city-years (2,000 resamples)
Ceiling test	Simulations of how well a perfect forecaster could do with data like ours, to separate model weakness from the limits of the data

Why AUC within each city, not accuracy: outbreak weeks are rare (about 7% of weeks), so a model that never raises an alert would be "93% accurate". AUC measures whether risky weeks are ranked above normal ones, and computing it within each city stops easy comparisons between a large city and a small one from flattering the score.

Validation results

The final model is the best forecaster we tested: it beats the seasonal calendar and our earlier model on all three main scores (log score, AUC and yearly error), 1–2 weeks ahead. It is still well below perfect, which we suspect is mostly because of the data.



Rolling-origin test and ceiling simulations (200 simulated histories), outbreak-excluded threshold

1 week ahead, 2014–2022	Seasonal calendar	Earlier model	**Final model**
AUC within city	0.67	0.73	**0.77**
Log score (lower is better)	0.898	0.832	**0.782**
Error in yearly case totals	97%	67%	**42%**
Alert at 10% chance: alerts that were real	17%	20%	**24%**
Alert at 10% chance: outbreak weeks caught	58%	62%	**71%**
Alert at 10% chance: false alarms per city per year	10.3	8.9	**7.7**
Alert at 20% chance: alerts that were real	21%	28%	**34%**
Alert at 20% chance: outbreak weeks caught	40%	48%	47%
Alert at 20% chance: false alarms per city per year	5.2	4.3	**3.2**

Each improvement is statistically clear (log-score gain per week, 95% CI by block bootstrap over 106 city-years):

- Adding rain to the seasonal calendar: +0.028 (0.007 to 0.050).
- Rain plus last week's cases: +0.066 (0.033 to 0.108) over the calendar.

- Final model over the earlier model: +0.049 (0.024 to 0.079). Most of this comes from using each city's recent case level instead of a long fitted trend.

By horizon: rain adds clear skill 1 and 2 weeks ahead, little at 3, and none at 4, matching the 2-week peak in risk. At 2 weeks ahead the final model's AUC is 0.76.

By city (final model, 1 week ahead):

City	Outbreak weeks	AUC within city	Forecast vs actual cases, 2014–2022
Mandaluyong	24	0.90	−30%
Dagupan	45	0.87	+44%
Iloilo	61	0.85	+22%
Muntinlupa	19	0.83	−7%
Navotas	39	0.79	−7%
Davao	87	0.71	0%
Zamboanga	44	0.70	−35%
Tacloban	15	0.68	+66%
Cagayan de Oro	29	0.62	−13%

Legazpi, Mandaue and Palayan had only 1–3 outbreak weeks each, too few to score. The model works best in the flood-prone Luzon and Visayas cities and worst in Mindanao, where heavy rain also explains the smallest share of cases. Cagayan de Oro used to be over-forecast by 133%: Typhoon Washi is 73% of its cases and taught the model a false yearly December–January peak. Setting Washi's weeks aside in training fixed it (now −13%, and forecasts better in all 9 test years), and the final model includes this fix.

What the ceiling test shows: a model fitted with hindsight on all years reaches 0.83, and a perfect forecaster that knew each week's true risk would reach 0.95. Even that perfect forecaster would misjudge weekly case counts by about 62%, and only about half its alerts at a 20% cut-off would be real. Small weekly counts set a hard limit, so the remaining gap needs new information, not more tuning.

Robustness checks:

- **Rainfall source:** training on research-grade IMERG Final and forecasting with real-time IMERG Late gives the same skill as Late throughout. The app can run on real-time data.
- **Reporting delay:** if cases were known the same week, the 1-week forecast would improve by more than rain itself adds. Faster PIDSR reporting would help more than better rain data.
- **Pooled vs regional rain curve:** no detectable difference (+0.002, CI −0.007 to 0.010), so the simpler pooled curve is used.

- **Probabilities:** in the earlier model's tests, forecast chances above about 20% ran too high (a "40%" week happened about 25% of the time). The app therefore shows risk levels, not percentages.

Risk Levels for the App

The app shows each city's risk 1 and 2 weeks ahead as **Normal**, **Watch** or **Warning**. On the unseen test years, Warning weeks became outbreak weeks 42% of the time, 6 times the usual rate.

How the levels are set: for each test year, the cut-offs come only from that year's training years. Warning = a forecast chance in the top 5% of training-year forecasts; Watch = the next 15%. One set of cut-offs is used for all cities.

1 week ahead, 2014–2022	Became outbreak weeks (95% CI)	vs usual rate	Share of all outbreak weeks	Weeks per city per year	False alarms per city per year
Warning	42% (37–48%)	6.1×	34%	2.9	1.7
Watch	16% (14–18%)	2.3×	37%	8.1	6.8
Normal	2.5% (2.1–3.1%)	0.4×	29%	39.7	—

- Watch and Warning together caught **71% of outbreak weeks**, and flagged the first week of **80% of multi-week outbreaks**.
- At 2 weeks ahead the levels hold up: Warning 38%, Watch 18%, Normal 2.7%. The 2014–2019 years (before COVID) give the same picture: Warning 41%, Watch 13%, Normal 2.5%.
- **Why one cut-off for all cities:** per-city cut-offs were tested and caught far fewer outbreaks (52% against 71%) for only a modest gain in reliability. Davao is flagged about 36 weeks a year with one cut-off, but it has the most outbreak weeks of any city, and those alerts catch 92% of them. The app emphasises Warning weeks there to limit alert fatigue.
- **Small cities** are rarely flagged because their risk is genuinely low; for them the app also shows a "confirmed unusual week" signal (2 or more cases).
- **Normal means low risk, not no risk:** 29% of outbreak weeks, mostly single-week blips, still fall in Normal weeks, so routine surveillance continues.

Suggested actions (to confirm with DOH guidance and local health offices): *Watch*, prepare: public advisories, alert clinics to fever after flood exposure, check doxycycline stock. *Warning*, act: targeted advisories in the most flood-exposed barangays, position doxycycline for prophylaxis, step up case finding.

Case Studies

LeapTo would have warned of the Navotas 2018 outbreak 2–3 weeks before case reports showed it, but would not have foreseen Typhoon Washi in Cagayan de Oro; Iloilo 2011–2012 is mixed. Each event was replayed two ways: **as of then** (trained only on earlier years and retrained each January, exactly as in the validation) and **today's model** (trained on all years except the event, so it never sees the outbreak).

Outbreak	Cases	Risk level for its first week (1 / 2 weeks ahead, as of then)	Lead over case reports	Outbreak weeks flagged (1 week ahead)
Navotas, 17–30 Jun 2018	8	Watch / Normal	2 weeks	2 of 2
Navotas, 8 Jul – 4 Aug 2018	19	Warning / Warning	2–3 weeks	4 of 4
Cagayan de Oro, Washi, 18 Dec 2011 – 21 Jan 2012	774	Normal in the surge and peak weeks; the earlier Watch was a generic low-threshold flag	None	Only after cases were reported
Iloilo, Jan–Feb 2011 (two outbreaks)	43	Warning / Warning	2–3 weeks	4 of 4
Iloilo, 7 Aug – 22 Oct 2011	795	Normal / Normal	None	8 of 11, from week 3
Iloilo, 13 Nov 2011 – 3 Mar 2012	281	Warning / Watch	2–3 weeks	14 of 16
Iloilo, 23 Sep – 17 Nov 2012	485	Normal / Watch	0–3 weeks	3 of 8

- **Navotas:** 11 of 12 outbreak weeks at Watch or Warning (9 at Warning), with 5 false-alarm weeks over 7 months. The calendar-only model flagged 4 of 12, none at Warning.
- **Washi:** the model expected about 0.3 cases a week during the storm. The surge week (123 cases) and the peak week (554) were both Normal; the level rose to Watch only after cases were reported. Rain over the city was not extreme (172 mm), the flooding came mostly from upstream, and the peak week was about 40 times the threshold.
- **Iloilo:** 71% of 48 outbreak weeks flagged as of then (85% with today's model), with 36 false-alarm weeks over 2 years. The largest outbreak was only flagged once reports would already have shown it.
- **For 2011–2012**, only the 3-year DOH threshold exists and the "as of then" model had about 2 years of usable data, so those results are a harder test than the app will face.

What it means: the app works for outbreaks that build over weeks after monsoon rain, the typical leptospirosis pattern. It cannot foresee a single catastrophic flood.

Recommendation: a separate Flood alert. After a typhoon or flood event with major flooding, the city health office switches on a **Flood alert lasting 4 weeks**. It appears alongside the forecast risk level, not in place of it, so the forecast stays exactly as validated: *Risk level: Normal (forecast) · Flood alert active, 3 weeks left.*

- **Why 4 weeks:** risk peaks about 2 weeks after heavy rain and stays raised for about 5 weeks, so 4 weeks from the flood covers the main danger window.
- **Why manual:** the people who know a major flood happened set it off. The automatic signals available here would have missed Washi: rain over the city was not extreme, and GloFAS ranked the flood only 20th in its year. An automatic trigger from PAGASA tropical cyclone warnings is future work.
- **Not part of the validation:** the Flood alert is a rule, not a model output. None of the accuracy figures in this document include it.

Flood Hazard

Flooding makes heavy rain more dangerous, and cases cluster where flood hazard is highest. But fixed flood-hazard maps do not make the weekly forecasts more accurate, so they are used in the app for targeting, not prediction.

Exposure measure: the share of each city's residents living in medium- or high-hazard areas on the UP NOAH 100-year flood map (the only map covering all 12 cities) (Thinking Machines Data Science, 2024). It ranges from 9% in Tacloban to 37% in Navotas.

1. Cases cluster in flood-prone barangays. Zamboanga has cases located to 101 barangays (328 cases, 2013–2022). Case rates rise **1.26× per 10 points** more flood-hazard area (95% CI 1.08–1.48), and 1.18× after allowing for population density. Barangays with more than 30% of their area in hazard zones had 2.5× the case rate of those under 10%. Muntinlupa's 9 barangays point the same way.

2. Heavy rain is more dangerous in flood-exposed cities. Adding flood exposure to the rain–risk model makes the rain effect **14.5% stronger per 10 points** of exposure (95% CI 4.8–23.9%, p = 0.005, AIC improves by 6).

City flood exposure	Peak-week risk after a heavy-rain week	Extra cases from that week
Low (9%, like Tacloban)	1.61×	2.0 typical weeks' worth
Average (23%)	1.81×	2.6
High (37%, like Navotas)	2.04×	3.3

The result held when each city was left out in turn (positive in all 12 runs, significant in 10). It appears only when exposure counts medium and high hazard where people live; low-hazard zones and area-based measures show no effect.

3. It does not improve the forecasts. Repeating the rolling test with flood-scaled rain (on the earlier model) left AUC (0.732 → 0.734), log score and alert precision essentially unchanged. A fixed map gives

each city one number, so it changes how alarming a city's heavy-rain weeks look, but not which of its weeks look riskiest.

How the app uses it: when a city's risk is raised, the app can point to its most flood-exposed barangays, using the barangay ranking from this analysis.

River Flow

Rising rivers raise leptospirosis risk beyond what city rain explains, but adding river flow does not improve the forecasts, so the final model stays as it is. River flow comes from GloFAS (daily, about 5 km grid, 2003–2022), which, unlike the flood maps, changes week to week (Copernicus Emergency Management Service, 2019).

Which rivers: each city's main river cell, fixed on a map before any results. Seven cities were used (Dagupan, Palayan, Navotas, Mandaluyong, Iloilo, Cagayan de Oro, Davao). Muntinlupa, Legazpi, Mandaue, Tacloban and Zamboanga were left out because their rivers are too small for the grid or their flooding comes from elsewhere; an all-12-cities version was run as a check. The weekly signal is the week's peak flow compared with that river's normal in 2003–2007, before any case data.

Check against known floods: Ondoy (2009) in Navotas and Mandaluyong, Pepeng (2009) in Dagupan and the 2012 Habagat rank among the highest flows of their year. Typhoon Washi in Cagayan de Oro ranks only 20th in 2011: a short flash flood that a 5 km daily model smooths out.

Test	Result
Explains cases, all years (river added on top of rain)	**Yes.** Fit improves by 68 AIC points ($p < 0.000001$); all four versions agree
Risk after a week in a river's top 5%	**1.17–1.25×** in each of the next 1–8 weeks; no effect in the same week
Overlap with rain	With river flow in, rain's peak-week effect falls from 1.77× to 1.52× and its total extra risk roughly halves
Forecast score, 1 week ahead (simple form)	+0.002 (95% CI −0.005 to +0.012): a tie; slightly worse at 2–4 weeks
Forecast score, 1 week ahead (lag-curve form)	−0.006: slightly worse, significantly so at 2–4 weeks
Warning weeks that became outbreaks	41.2% without river, 40.6% with (outbreak weeks caught 70.5% vs 70.8%)
Typhoon Washi, Cagayan de Oro	774 cases reported; 43 expected with rain only, 42 with river flow

Why it explains but does not forecast: by forecast time the model already knows the recent rain and the recent case level. A river that was high 1–4 weeks ago has mostly already shown up in those, so little

new is left. The all-12 version looked slightly better on AUC, but that came from Tacloban, where it also over-forecast yearly cases by 258%.

What it adds to the story: together with the flood-hazard result, it points to **flooding as the pathway from rain to cases**. Part of what the model credits to rain works through rivers rising. Rain over each river's upstream basin was not tested separately: river flow already carries that rain, and an earlier check of Cagayan de Oro's upstream catchment found no gain over city rain.

Limitations

The model is a risk-warning tool, not a case counter, and its evidence has known gaps.

- **Small counts:** most city-weeks have 0–2 cases, so weekly case numbers are dominated by chance. Even a perfect forecaster would miss them by about 60%.
- **Many false alarms at sensitive cut-offs:** catching 70% of outbreak weeks costs about 8 false alarms per city per year. A stricter cut-off cuts false alarms but misses more outbreaks.
- **Short warning window:** risk peaks 2 weeks after rain, so warnings come 1–2 weeks ahead and often in an outbreak's first or second week, not well before it.
- **Uneven by city:** weak in Zamboanga, Tacloban and Cagayan de Oro; the three smallest cities have too few cases to judge.
- **Cagayan de Oro over-forecast**: fixed. Typhoon Washi, 73% of the city's cases, made the model expect a big outbreak every December–January; setting it aside in training cut the error from +133% to −13%.
- **Reporting:** PIDSR counts depend on people seeking care and being reported; 2020–2022 reporting was disrupted by COVID.
- **Limited history:** 15 years and 370 outbreak weeks in the test years, so per-city results carry wide uncertainty.
- **Flood maps:** one 2015 snapshot of river and rain flooding only (no storm surge or tides), with people assumed spread evenly within barangays.
- **Flash floods:** extreme, short floods like Typhoon Washi are underestimated by both satellite rain and the 5 km river model, so the model expected about 43 cases where 774 were reported, and the app would not have warned before Washi (see Case studies).
- **Week calendar:** the exact DOH week start could not be fully confirmed; the main findings held across three calendar versions.

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